# Engineering Tools

*Essential Digital Tools & Utilities for Modern Engineers*

## What Are JC Tech Engineering Tools?

JC Tech Engineering Tools is a curated collection of free digital utilities designed to solve everyday engineering problems quickly and accurately. From unit converters and material selectors to stress calculators and tolerance stack-up analyzers, these tools eliminate manual calculations and reduce design errors. Every tool is validated against textbook solutions and industry-standard software.

## Tool Categories

- Unit Conversion Suite: Comprehensive converter covering SI, Imperial, and engineering-specific units (torque, pressure, flow rate, hardness).
- Material Property Lookup: Searchable database of mechanical properties (yield strength, UTS, modulus, density, CTE) for 200+ materials.
- Stress & Deflection Calculators: Beam bending, torsion, column buckling, pressure vessel, and contact stress calculations.
- Tolerance Stack-Up Analyzer: Worst-case and statistical (RSS) tolerance analysis for multi-part assemblies.
- Gear Design Calculator: Spur, helical, and bevel gear sizing with Lewis beam strength and AGMA methodology.
- Fastener Selection Guide: Bolt grade selection, torque-tension calculator, and thread engagement depth validator.

## How It Works

Each tool provides a clean input interface where you enter your known values. The tool performs validated calculations and displays results with full formula breakdowns, unit labels, and applicable standard references.

Results can be exported as PDF reports for documentation.

## Featured Tool: Beam Deflection Calculator

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| **Supported Beams:** | Simply supported, cantilever, fixed-fixed, overhanging, continuous |
| **Load Types:** | Point load, UDL, triangular, moment, combined loading |
| **Outputs:** | Max deflection, slope, bending moment diagram, shear force diagram |
| **Standards:** | Validated against Timoshenko & Gere, Roark's Formulas for Stress & Strain |
| **Export:** | PDF report with diagrams and calculation steps |

## Featured Tool: Tolerance Stack-Up Analyzer

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|---|---|
| **Methods:** | Worst-case arithmetic, Root Sum of Squares (RSS), Monte Carlo simulation |
| **Inputs:** | Nominal dimension, tolerance, distribution type per component |
| **Outputs:** | Assembly gap/interference, Cpk prediction, probability of fit |
| **Use Case:** | Multi-part assembly validation before prototyping |

## Upcoming Tools (Roadmap)

- Weld Strength Calculator (AWS D1.1 compliant) -- Q1 2027
- Thermal Expansion Interference Fit Calculator -- Q1 2027
- Fatigue Life Estimator (S-N curve and Miner's Rule) -- Q2 2027
- Hydraulic Cylinder Sizing Tool -- Q2 2027
- Spring Design Calculator (compression, extension, torsion) -- Q3 2027

## Access the Tools

All Engineering Tools are available free of charge through the JC Tech Knowledge Hub. No login required for basic tools; advanced features available with a free JC Tech account. Enterprise API access is available for companies wanting to embed these calculators into their internal engineering portals.

Visit: www.jctech.co.in/tools | Email: tools@jctech.co.in